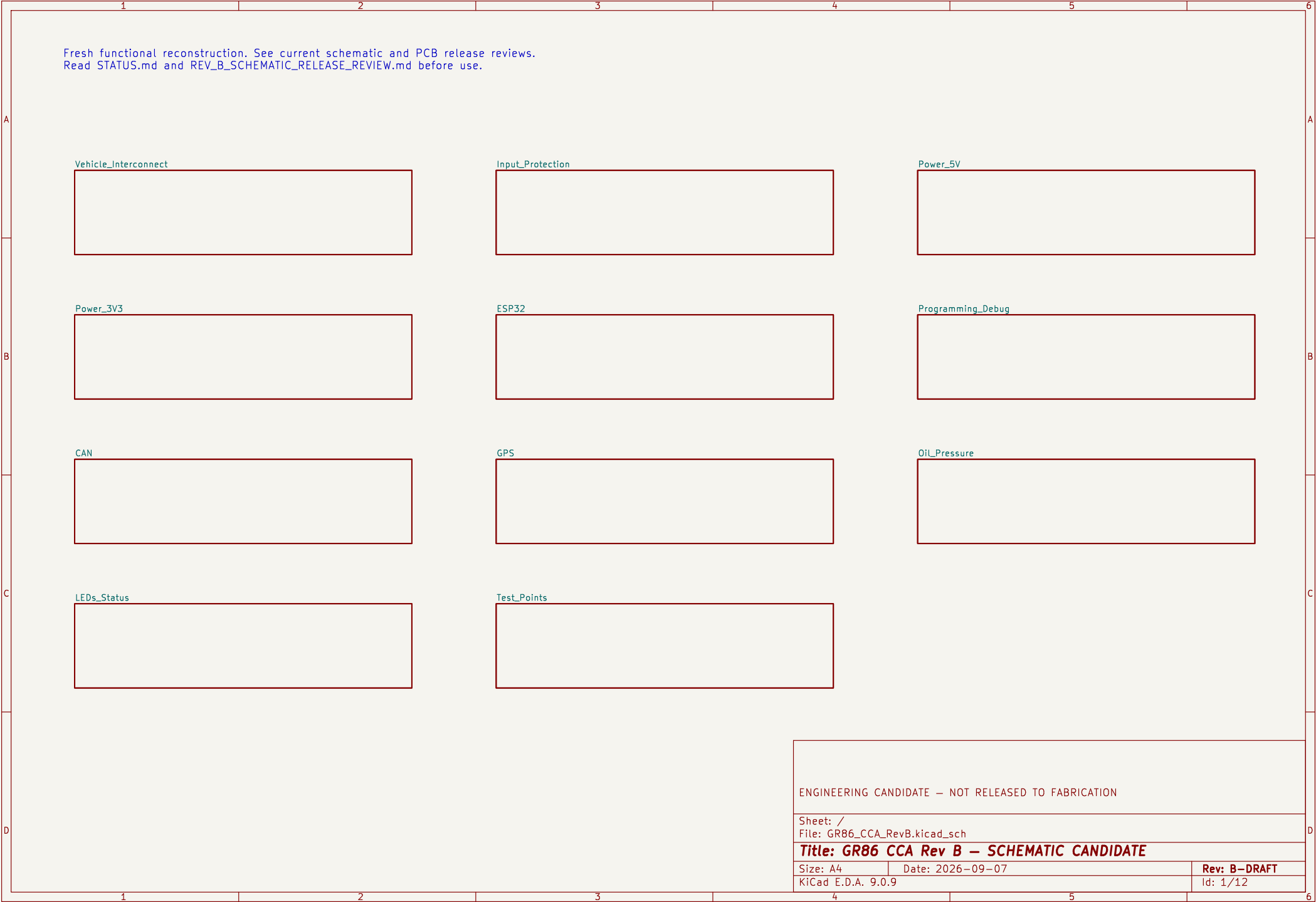
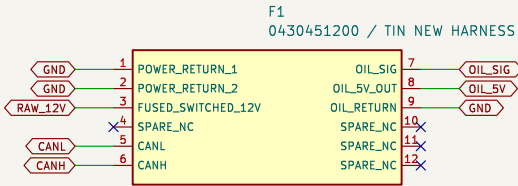




RVB-01 CONTROLLED HARNESS CONTRACT
Header0430451200 TIN; housing0430251200.
TIN terminals0430300007 loose /0430300001 reel.
Physical cavity numbers follow Molex. Keying and wire colors do not redefine pin numbers.



1 return | 2 HARNESS EMPTY (PCB GND) | 3 fused switched12V | 4 NC
5 CANL | 6 CANH | 7 oil signal | 8 protected5V OUTPUT | 9 oil return
10/11/12 NC. Seven leads; cavities2/4/10/11/12 empty. No IGN/chassis straps.
INT22-02: one full-current return atf1.1; no parallel-load current sharing.
F1 wire exit relaxed25.4-30mm before bend/tie/twist; tin-to-tin mating only.
ASC cavity map, actual plug exit, crimp and continuity remain installation gates.
Use current HARNESS_INTERFACE_CONTRACT; keyed face/latch orientation must be bound.
Oil return joins the continuous PCB reference; no sensor-A-to-case jumper.
Pin2 stays electricallyGND in CAD; the harness terminal/wire is omitted.

ENGINEERING CANDIDATE – see release reviews

GR86 / BRZ telemetry

Sheet: /Vehicle_Interconnect/
File: Vehicle_Interconnect.kicad_sch

Title: Vehicle_Interconnect – SCHEMATIC CANDIDATE

Size: A4

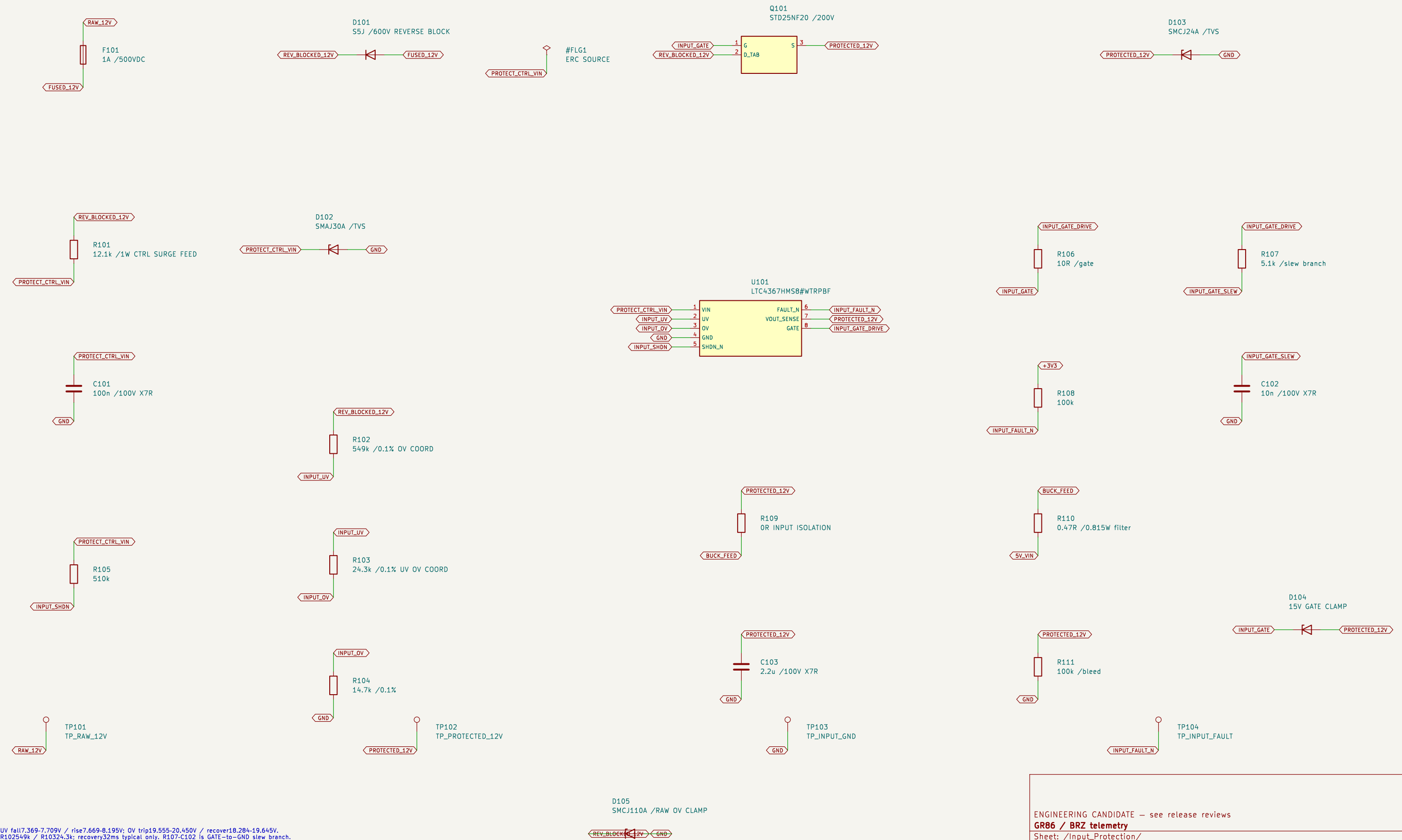
Date: 2026-09-07

Rev: B-DRAFT

KiCad E.D.A. 9.0.9

Id: 2/12

RVB22 – REVERSE BATTERY + OVERVOLTAGE DISCONNECT
Normal9.5-16V;24V is disconnect/survival, not normal regulation.
Thresholds are sensed after input diode at R102.1 / REV_BLOCKED_12V.
See RVB22_BRING_UP_AND_MEASUREMENT_CONTRACT.md; no ISO pulse pass is implied.



UV fall7.369-7.709V / rise7.669-8.195V; OV trip19.555-20.450V / recover18.284-19.645V.
R102549k / R10324.3k; recovery32ms typical only. R107-C102 is GATE-to-GND slew branch.
R109 open isolates converters; begin20mA limit; investigate protection-only current>0.6mA at16V.
Controller VIN has its own current-limited clamp; D103 is downstream of disconnect.

ENGINEERING CANDIDATE – see release reviews

GR86 / BRZ telemetry

Sheet: /Input_Protection/

File: Input_Protection.kicad_sch

Title: Input_Protection – SCHEMATIC CANDIDATE

Size: A3

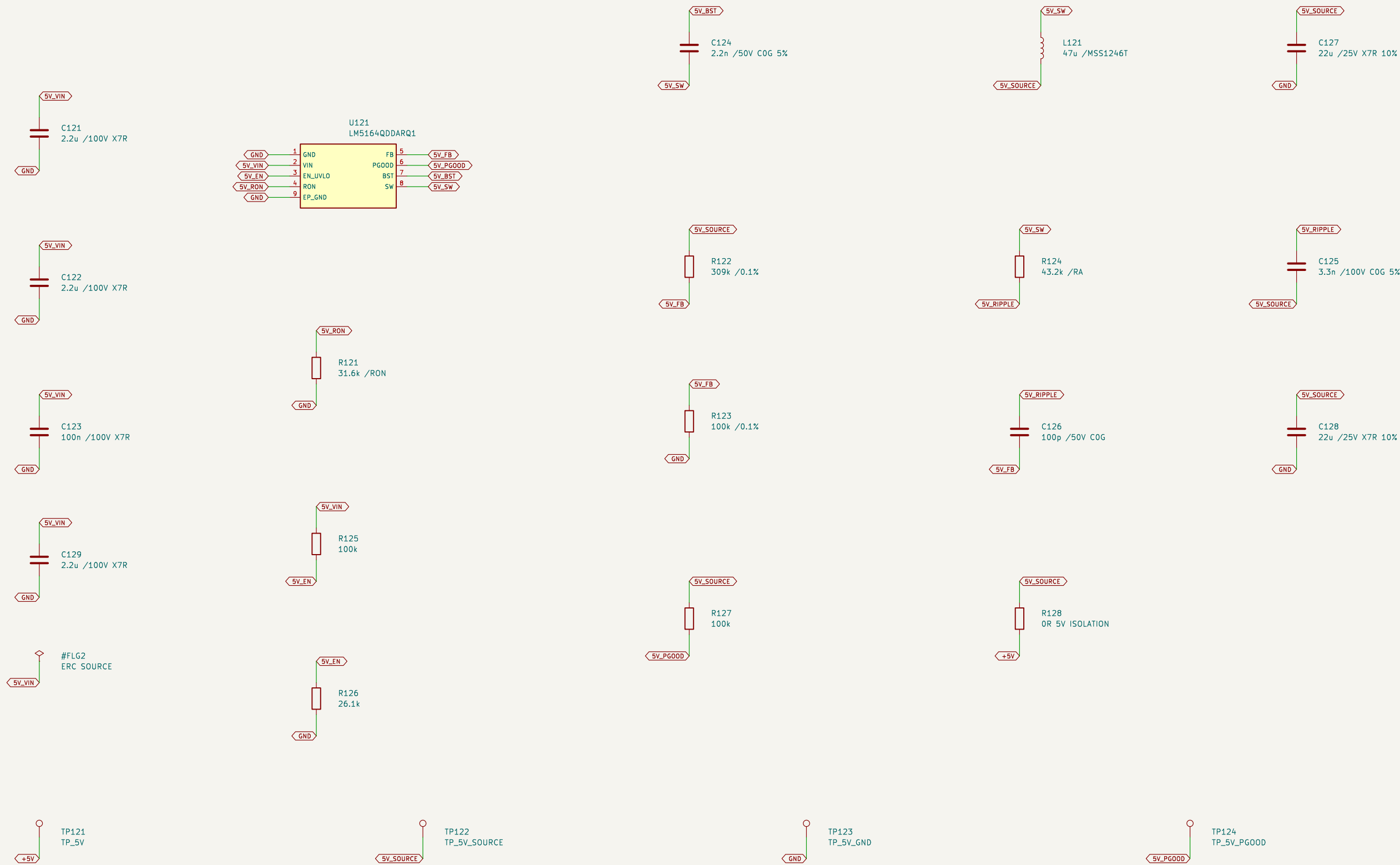
Date: 2026-09-07

Rev: B-DRAFT

KiCad E.D.A. 9.0.9

Id: 3/12

RVB-02 – LM5164-Q1 WIDE INPUT BUCK /5V /1A CAPABILITY
Fresh physical pin map; COT TYPE3 ripple injection is required.
Nominal rail target5.0V; feedback accounts for ripple offset. See calculations.



R125/R126 gives local start-7.25V / stop-6.76V; upstream UV disconnect acts first in steady state.
2.2nF bootstrap value is intentional, CA is between RIPPLE and VOUT; CB couples to FB.
Kelvin FB bottom return to pin1/EP. Minimize VIN-SW-GND hot loop and switch-node copper.
L121 manufacturer recommended land pattern is derived in POWER_FOOTPRINT_REVIEW.md.

ENGINEERING CANDIDATE – see release reviews

GR86 / BRZ telemetry

Sheet: /Power_5V/
File: Power_5V.kicad_sch

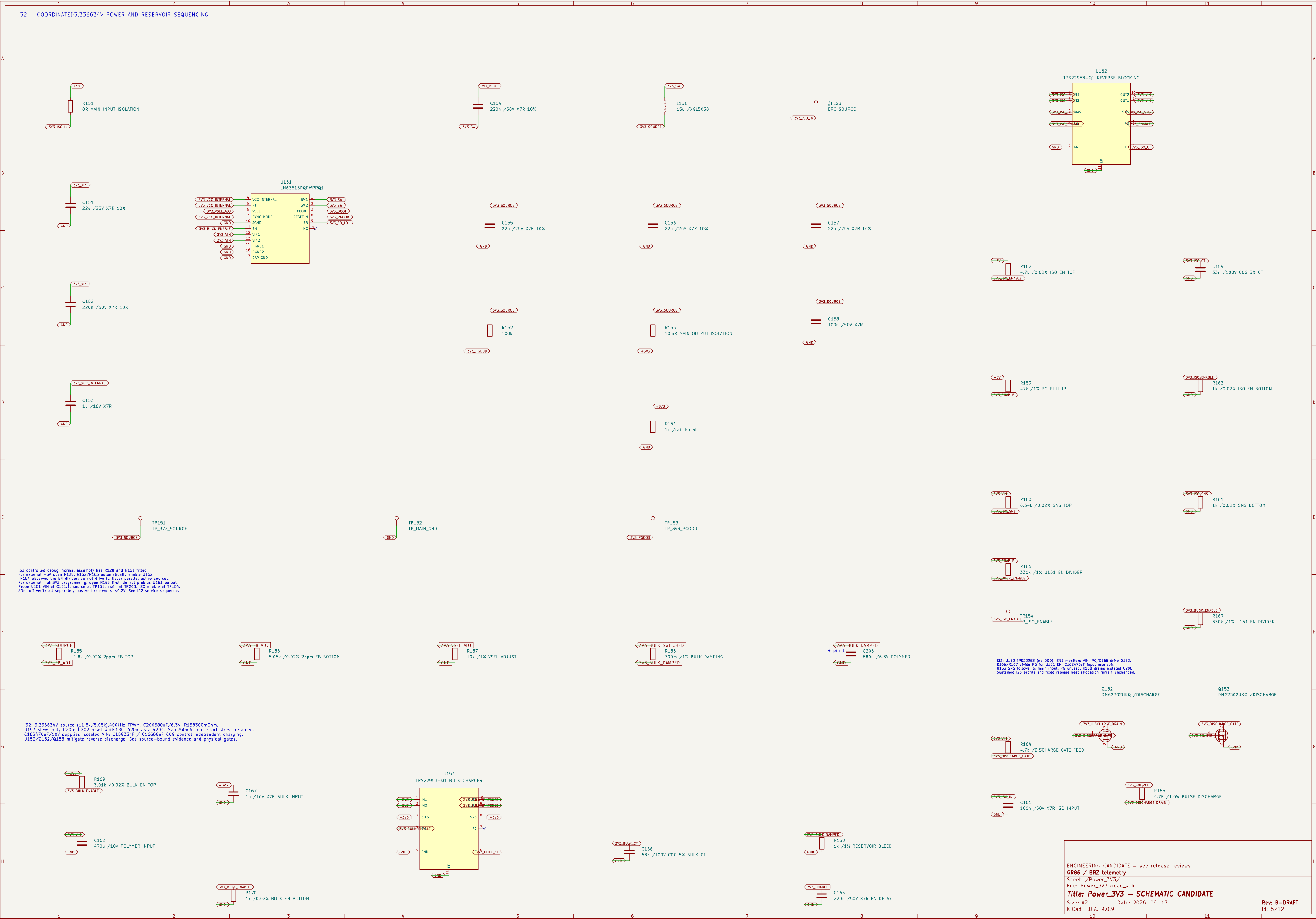
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Size: A3
KiCad E.D.A. 9.0.9

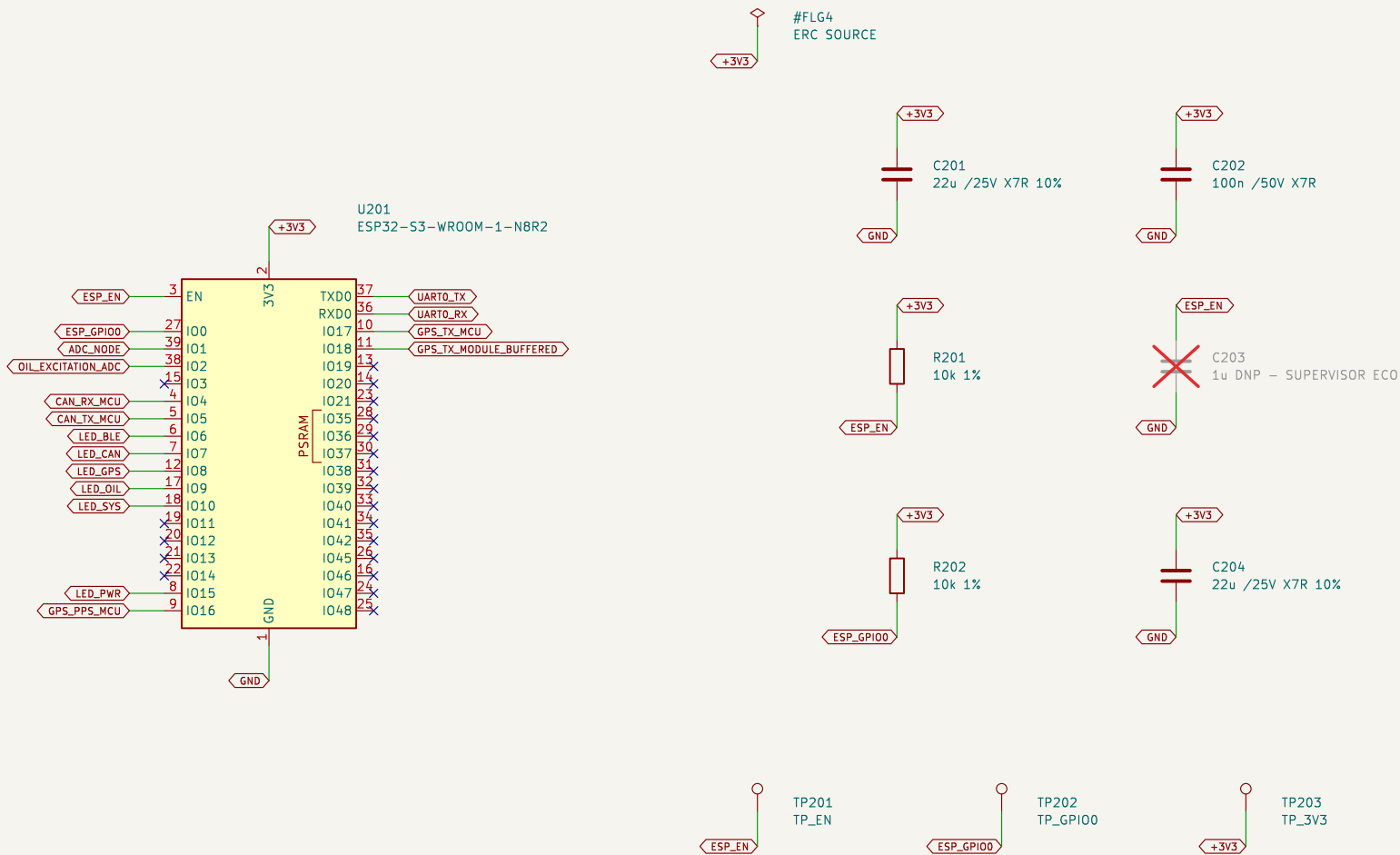
Date: 2026-09-07

Rev: B-DRAFT

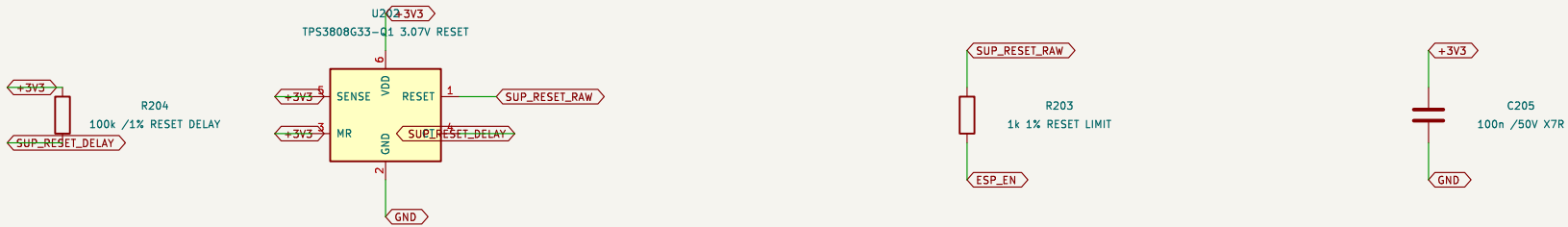
Id: 4/12



RVB-03 WORKING CIRCUIT – independent official module pin map
N8R2 temperature grade chosen; antenna/thermal qualification remains HOLD.



GPIO3 / GPIO46 are strapping pins and intentionally NC. OIL LED GPIO9; PWR LED GPIO15.
GPIO35-37 unused. GPIO2 excitation monitoring is a source implementation; target build remains required.
Firmware2.1.4 source and calibration/invalid-data contract require current target verification.
RVB22 supervisor: R201 pull-up retained; C203 DNP. No current compiled/native pass is claimed.



I32 RESET: U202 falling3.023950–3.116050V; release threshold max3.193952V.
R204100k ties CT toVDD; published180–420ms delay permits C206 charging before boot.
C203 remains DNP. Normal rail transients must stay>3.116050V and<3.6V without reset.
Retain programmer open-drain reset; hold GPIO0 until actual EN release.
Physical startup/boot/reset and RevA fault correlation remain acceptance gates.

ENGINEERING CANDIDATE – see release reviews

GR86 / BRZ telemetry

Sheet: /ESP32/

File: ESP32.kicad_sch

Title: ESP32 – SCHEMATIC CANDIDATE

Size: A3

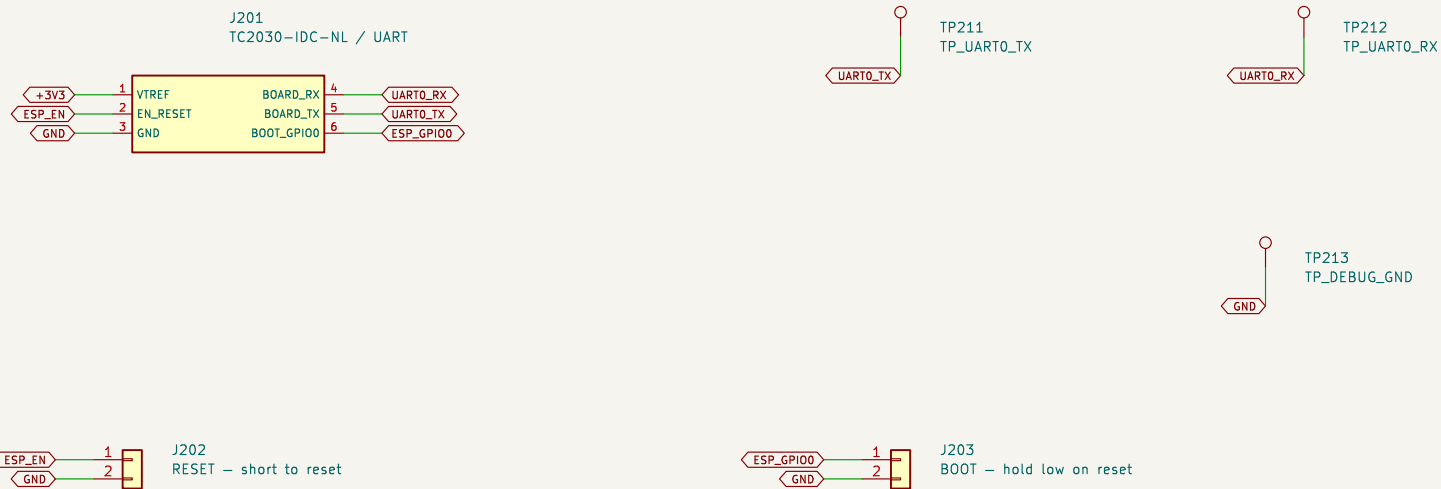
Date: 2026-09-07

Rev: B-DRAFT

KiCad E.D.A. 9.0.9

Id: 6/12

RVB-03 – CUSTOM UART ADAPTER CONTRACT
TC2030-IDC-NL, top-side contact. Not a Microchip ICD connection.



1 VTREF (sense only) 2 EN/RESET (programmer open drain) 3 GND
4 BOARD RX <- programmer TX 5 BOARD TX -> programmer RX 6 GPIO0/BOOT
Only 3.3 V UART levels. No power injection at VTREF. Cable pin n maps to IDC pin n.
Pin-1 triangle and no-paste contact pads are mandatory. Cable/clip access is a placement gate.
Boot: ground GPIO0; hold actual EN low >=50us; release EN; keep GPIO0 low >=3ms after actual EN is high and until ROM download starts. RVB21 supervisor holds EN low 12-28ms after rail recovery.
J202/J203: Wurth61300211121. Adapter continuity and mechanical access remain OPEN.

ENGINEERING CANDIDATE – see release reviews		
GR86 / BRZ telemetry		
Sheet: /Programming_Debug/		
File: Programming_Debug.kicad_sch		
Title: Programming_Debug – SCHEMATIC CANDIDATE		
Size: A4	Date: 2026-09-07	Rev: B-DRAFT
KiCad E.D.A. 9.0.9		Id: 7/12

The schematic shows the electrical connections for a CAN bus system. The central component is the TCAN3403DRBRQ1 (U301), which interfaces between the MCU (CAN_TX_MCU, CAN_RX_MCU) and the vehicle CAN bus (CANH, CANL). It includes a +3V3 supply, a VIO pin, and a STB pin. A 1kΩ resistor (R302) connects the CAN_RX_MCU to the RXD pin. A 10kΩ resistor (R303) holds the TXD pin recessive. An ESD protection diode (D301, ESD2CAN24DBZRQ1) protects the CANH/CANL pins from transients. Termination is provided by a 120Ω resistor (R306) at the vehicle end, controlled by the SJ301 bench terminal switch. Decoupling capacitors C301 (100nF /50V X7R) and C302 (1μF /16V X7R) are used for power filtering. Test points TP301 through TP305 are located along the signal and ground paths.

Pin 5 is VIO on TCAN3403; it is NOT VREF or SHDN. Pin 8 STB is hard low for full-bandwidth RX. Standby mode is not used as a silent receive mode. External R303 holds TXD recessive. Vehicle firmware must use listen-only. Isolated bench ACK testing needs R301 fitted and normal mode. CCA receives passively with R301 open; a separate known-good bench node must ACK the transmitter. Bus ESD device is not proof of vehicle transient compliance. Connector/ground release remains HOLD.

ENGINEERING CANDIDATE — see release reviews		
GR86 / BRZ telemetry		
Sheet: /CAN/		
File: CAN.kicad_sch		
Title: CAN — SCHEMATIC CANDIDATE		
Size: A4	Date: 2026-09-07	Rev: B-DRAFT
KiCad E.D.A. 9.0.9	Id: 8/12	

Pin 5 is VIO on TCAN3403; it is NOT VREF or SHDN. Pin 8 STB is hard low for full-bandwidth RX. Standby mode is not used as a silent receive mode. External R303 holds TXD recessive. Vehicle firmware must use listen-only. Isolated bench ACK testing needs R301 fitted and normal mode. CCA receives passively with R301 open; a separate known-good bench mode must ACK the transmitter. Bus ESD device is not proof of vehicle transient compliance. Connector/ground release remains HOLD.

GR86 / BRZ telemetry

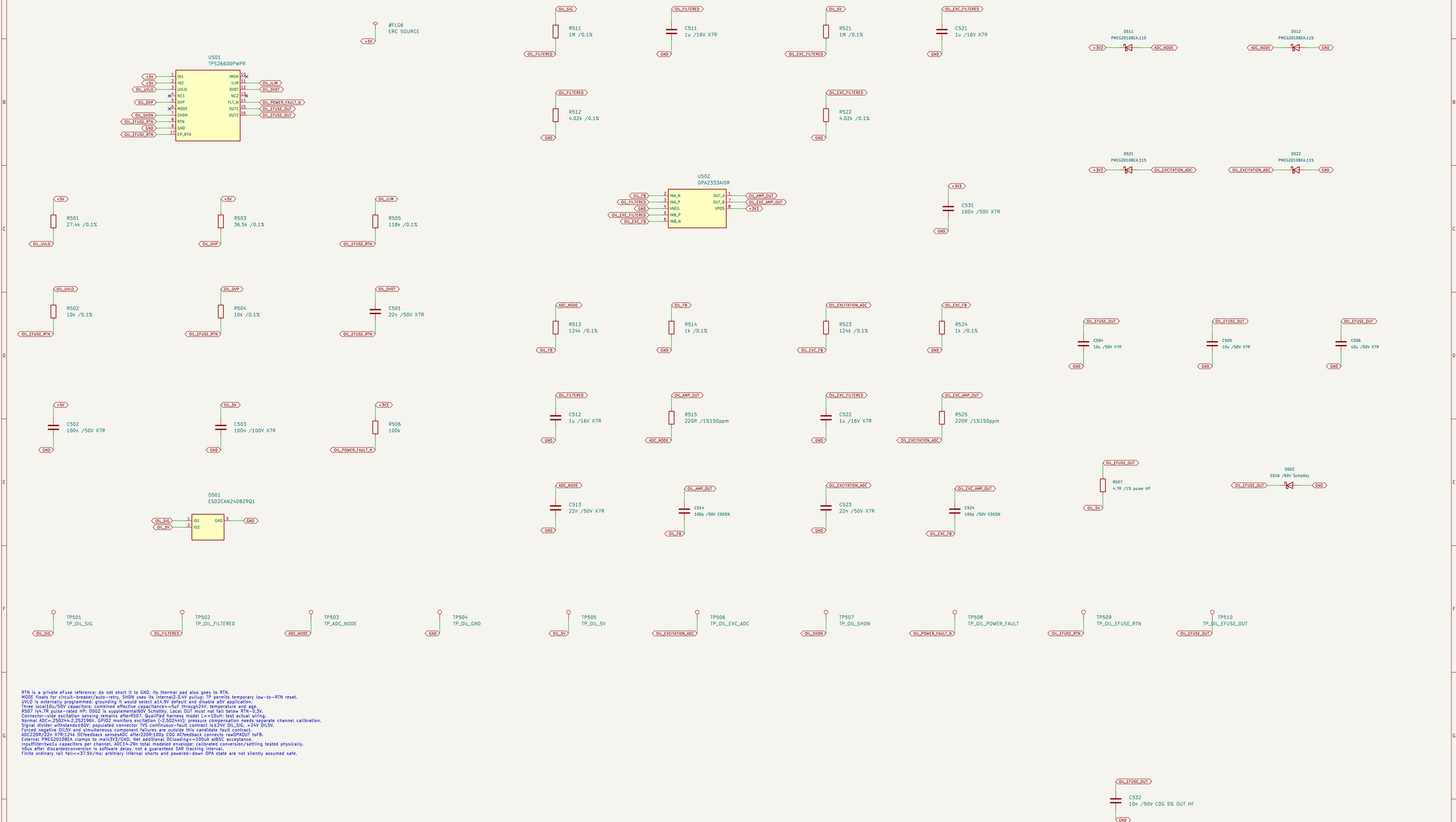
Title: CAN – SCHEMATIC CANDIDATE

Rev: B-DRAFT

Id: 8/12

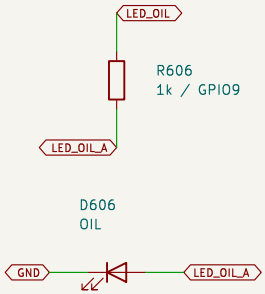
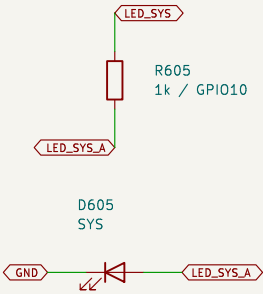
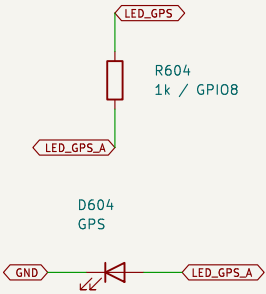
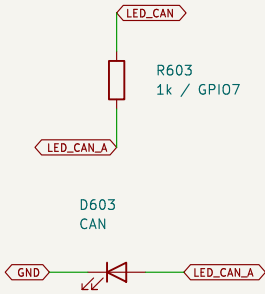
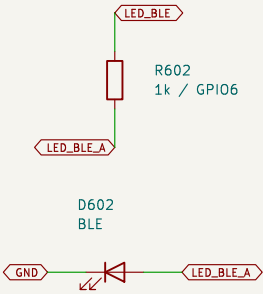
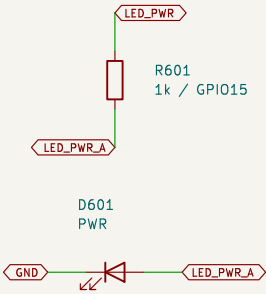
[illegible][illegible]

RVB-06 – HONEYWELL150psi /RATIOMETRIC5V /GPIO1 ADC
Connector signal → 1:249.76 attenuator → precision gain125 → low-impedance ADC drive.
OIL5V excitation is current limited and reverse-current blocked; no direct5V-to-ADC path.



RVB22 C532 OUT—to—GND HF bypass:10nF C0G50V; never privateRTN.
Direct U501 OUTpadding feed also joins C504 OUTland and existing in1 OUTcopper; use actual non—tree nodal model.
Preserve C504/C505/C506 bulk and C514 feedback. New—loop25nH/common—star10nH are allocations; native fill and hardware remain unexecuted.

STATUS INDICATORS — ACTIVE HIGH
GPIO3/GPIO46 strapping pins are free of external LED loads in Rev B.
Firmware Rev B contract moves OIL to GPIO9 and PWR to GPIO15.



1000ohm current limiting. See qualification/LED_STRAP_REVIEW.md.
Indicators represent firmware state and do not replace physical rail/bus measurements.

ENGINEERING CANDIDATE — see release reviews

GR86 / BRZ telemetry

Sheet: /LEDs_Status/
File: LEDs_Status.kicad_sch

Title: LEDs_Status — SCHEMATIC CANDIDATE

Size: A4

Date: 2026-09-07

Rev: B-DRAFT

KiCad E.D.A. 9.0.9

Id: 11/12

	1	2	3	4	5	6
A	TEST AND ISOLATION DIRECTORY Probe points are placed on the functional sheets beside their circuits. Bring-up removes rail0ohm links before testing each source independently. R301 TXD and R306 termination are DNP; SJ301 is open for vehicle tap use. L406 RF matching inductor is factory fitted in RVB21 candidate for the specified active antenna. RF/ESD/cable acceptance remains OPEN. See BRING_UP_PLAN.md and the machine-checked complete testpoint inventory. Do not confuse bare probe pads/Tag-Connect landing with purchased assembly parts.					A
B						B
C						C
D	ENGINEERING CANDIDATE — see release reviews GR86 / BRZ telemetry Sheet: /Test_Points/ File: Test_Points.kicad_sch Title: Test_Points — SCHEMATIC CANDIDATE Size: A4 Date: 2026-09-07 Rev: B-DRAFT KiCad E.D.A. 9.0.9 Id: 12/12					D
	1	2	3	4	5	6